

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1341

Van Buren 138 kV
November 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1341. The Interconnection Customer proposes a 167.5 MW battery energy storage facility interconnecting at the Van Buren 138 kV substation in the Pennsylvania area. The request was received on November 25, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 501 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,506,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Zachary – Levee Substation 138 KV
Contingency: Talenenergy Martins Crk Llc Harrisburg Substation 500-345 kV TX
Loading: 124.1% (Rating: 215 MVA)
DFAX: 5.5%
Pre-project loading: 119.9%
2. Facility: West Point – Powell 138 KV Line
Contingency: West Hempfield – O'Fallon Renewable Energy Ctr 138 KV
Loading: 110.3% (Rating: 290 MVA)
DFAX: 31.0%
Pre-project loading: 92.4%
3. Facility: Van Buren Substation – Powell 138 KV line
Contingency: Heritage Garden Wind Farm I Llc Substation – Van Buren 138 KV
Loading: 107.5% (Rating: 184 MVA)
DFAX: 100.0%
Pre-project loading: 16.4%
4. Facility: Van Buren – Heritage Garden Wind Farm I Llc 138 KV
Contingency: Van Buren – Powell Substation 138 KV L
Loading: 103.5% (Rating: 191 MVA)
DFAX: 100.0%
Pre-project loading: 15.8%
5. Facility: Ipl Ft Monrovia I Llc – Crane'S Mill 138 KV line
Contingency: O'Fallon Renewable Energy Ctr – West Hempfield 138 KV
Loading: 102.1% (Rating: 547 MVA)
DFAX: 15.2%
Pre-project loading: 97.5%

6. Facility: Crane'S Mill – Culebra 3801 138 KV L
 Contingency: O'Fallon Renewable Energy Center – W Hempfield 138 KV L
 Loading: 101.8% (Rating: 557 MVA)
 DFAX: 15.2%
 Pre-project loading: 97.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Zachary – Levee Substation 138 KV	Rebuild line with higher-capacity conductor	\$19,095,000	100% (\$19,095,000)
West Point – Powell 138 KV Line	Rebuild line with higher-capacity conductor	\$11,112,000	19% (\$2,089,000)
Van Buren Substation – Powell 138 KV line	Reconductor line	\$1,418,000	100% (\$1,418,000)
Van Buren – Heritage Garden Wind Farm I Llc 138 KV	Reconductor line	\$3,655,000	100% (\$3,655,000)
Ipl Ft Monrovia I Llc – Crane'S Mill 138 KV line	Reconductor line	\$1,511,000	4% (\$68,000)
Culebra 3801 – Crane'S Mill 138 KV	Reconductor line	\$4,038,000	4% (\$181,000)
	Total allocated to project		\$26,506,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.